

Interactive Data Science and Visualization: Multi-Disciplinary Open Educational Resources for Building Modern Data Experiences

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Software

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Summary

These open educational resources offer hands-on instruction in crafting digital experiences enabling users to interactively engage with data and computation through visualization. Taught at the University of California, Berkeley (Pottinger & Pérez, 2025) and in guest lectures / workshops across multiple schools and programs (Pottinger, 2026), these lessons begin with traditional information design (Munzner, 2014) and perception science (Ware, 2020). Then, instruction extends to media studies (Hall et al., 1980), human-centered computing (Harrison et al., 2007), user research (Shneiderman & Plaisant, 2006), and game design (Schell, 2019). Meanwhile, skills labs and guided practical projects explore the software engineering aspects of this multidisciplinary work, inviting scientists, engineers, journalists, designers, and others with programming foundation to build within applied exercises utilizing the unique possibilities of interactive digital media for understanding data (Portnow et al., 2012; Victor, 2012). Historically situated (Bitzer & Johnson, 1971) but embracing modern tools like AI, these reusable resources support instructors and learners in creating “media for thinking the unthinkable” (Victor, 2013).

Statement of Need

Through hands-on projects, these open source educational materials aim to empower students to critically design interactive digital media for interrogating data. However, going beyond traditional introductory information design instruction, these resources also explore the implementation of these rich bespoke interactive computational experiences as made possible through custom programming¹. Altogether, this educational contribution situates design techniques within engineering instruction as necessary for building media for thought (Victor, 2013).

State of the field

Existing open data visualization instruction often takes a perception science (Stahmer et al., 2026) and user-centered design approach (Munzner, 2021). These materials may primarily focus on reusable charts or static drawing (Cairo & Collaborators, 2026). While these valuable resources offer a robust introduction to traditional information design, many existing options choose not to center creating new bespoke experiences possible only with custom programming (Marcus & Wu, 2012).

¹Taught in Python, adaptable to other languages.

Learning objectives

In response to current open offerings and experience lecturing within existing curricula (Pottinger, 2026), the presented educational materials instead assume some programming capability enabling the creation of new custom interactive graphical forms. Honoring classical principles (Cleveland & McGill, 1984) but considering them flexibly in novel structures (Pottinger, 2024), this approach reaches beyond standard charts towards giving students the capacity to build pieces similar to the custom interactive experiences which often light up the classroom but which also often remain out of reach for students to actually construct (Kamvar & Harris, 2011; Rees & Periscopic, 2013). This embrace of software engineering to augment and reach new design topics enables teachers and learners to update traditional data visualization instruction such as in supporting science-informed policymaking (Alfredo et al., 2024; Pottinger et al., 2023) and interactive science (Victor, 2011), meeting new opportunities from computers' ever-growing capacity to react to the user (Victor, 2012). In total, these adaptable modular resources seek to empower learning blending evidence-based design and programming's expressiveness.

Design

In addition to discussing ethics and accessibility, modular resources explore different perspectives within data visualization. These lessons and projects are designed to be completed with minimal local software configuration.

Instructional design

The course is organized by ways of understanding data visualization and interactive data science:

- *Representation*: traditional perception science approach to information design (Ware, 2020).
- *Task*: modern user-centered design (Harrison et al., 2007; Munzner, 2014).
- *Message*: consideration of sociology, anthropology, and media studies (Hall et al., 1980; Tong, 2025).
- *Dialogue*: unique capabilities of digital interactivity (Victor, 2012) with audiences as co-creators of meaning, including through game design (Portnow et al., 2012; Pottinger, 2023).

Content

This mix of traditional lecture content with hands-on labs and projects² also often include supplementary captioned videos and PDFs. Altogether, these offer the experience of programming to not just learn interactive information design but to feel and understand the role of human iteration. This set of lessons concludes with critical but applied consideration of AI in visualization.

Software design

Supporting blended design and engineering instruction, materials are compiled from markdown and YAML to HTML at <https://mooc.interactivedatascience.courses/> via Python. All instructional materials are available under a Creative Commons license with open source code under a BSD-3-Clause license. Using modern technologies like WebAssembly, one may complete all instruction without installing local software beyond a modern web browser.

²Each entry includes its own citations which may extend beyond those listed here.

Background

These materials were originally developed in a university setting across multiple courses.

Origin

These materials were taught as a full course at the University of California, Berkeley in 2025 (Pottinger & Pérez, 2025). It also served different programs as components in other instruction or as workshops (Pottinger, 2026). These lectures have spanned multiple universities with inclusion in data science, design, and engineering contexts.

Future use

Individual lessons may provide topic-specific supplement to other instruction, modules may offer a treatment of a topical area for learning, and the class as a whole may allow for an approximate semester of hybrid hands-on and lecture-based exploration. The original full course was for 2 credit hours upper-division undergraduate or graduate. However, use of some optional materials may allow for use in a 3 credit hour setting.

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AI usage disclosure

Carefully directed AI aided in the HTML rendering system (Anthropic, 2024). With the exception of the AI skills lab which generated some content based on demo chat history, all written materials including lessons and labs were first written manually. They may have been proofread by AI but, unless trivial “mechanical” edits, AI-assisted changes (including proofreading and formatting) marked in git through co-author.

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